L-bracket: HOLLOW DESIGN ACCEPTED

```
cshell design accepted: J = +0.0721 (optimisation grid), +0.0731 (shifted grid) > ln(1 + g_min) = 0.0198,
feasible, export mass-matched and watertight

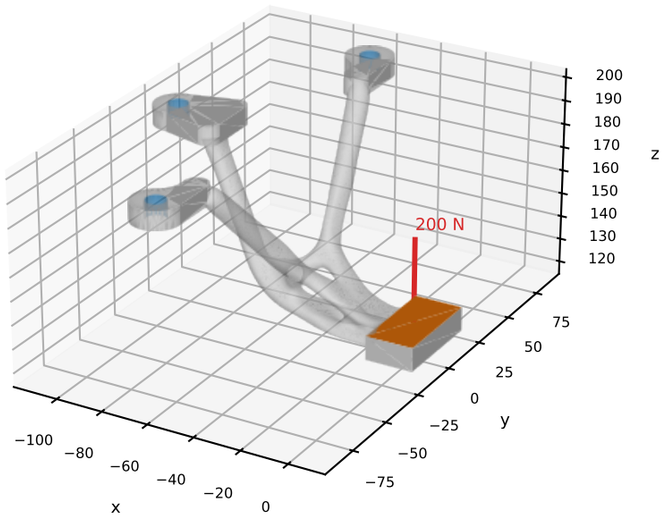
mass      target 43.63 g   final 43.63 g   error 5.5325e-06
optimiser grid  J 0.0721 (weighted)  ln K/K0 0.09453  K/K0 1.099  P/P0 1.051  governing strength  feasible True   (accept: J
    > ln 1.02 = 0.0198, K/K0 >= 1, P/P0 >= 1, sigPN0/sigPN >= 1, lamB/lamB0 >= 1)
shifted grid    J 0.07309 (weighted)  ln K/K0 0.0945  K/K0 1.099  P/P0 1.053  governing strength  feasible True   (accept: J
    > ln 1.02 = 0.0198, K/K0 >= 1, P/P0 >= 1, sigPN0/sigPN >= 1, lamB/lamB0 >= 1)

export    h 0.2 mm  faces 1599944  watertight True  volume 4.195e+04 mm3  material bodies 1  shells 1  enclosed voids 2
    escape holes 4
wall [mm] min 0.4863  p05 1.514  median 5.184  max 42.21
closed_area_frac   0.4332
hollow_area_frac   0.2818

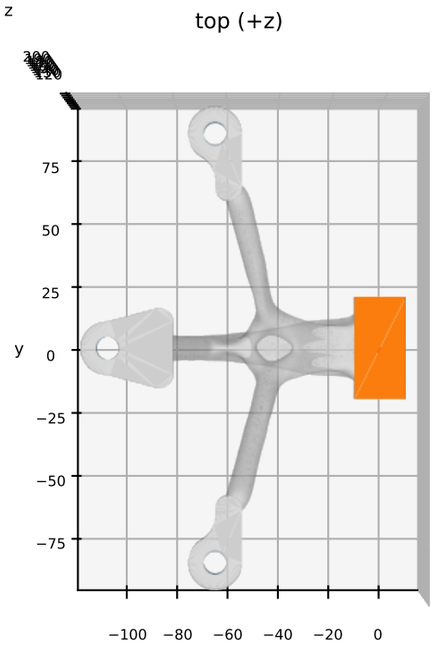
optimiser: {'method': 'mma', 'starts': ['solid'], 'max_iter': [60, 80, 40], 'move': 0.1, 'tol_x': 0.005, 'tol_f': 0.0001,
    'mass_tol': 0.0001, 'sliver_weight': [0.0, [0.0, 0.05], 0.1], 'seed': 0}
grid: {'phases': [{'h_e': 0.8, 'h_g': 0.2}, {'h_e': 0.4, 'h_g': 0.1}], 'export_h_mm': 0.05, 'rho_min': 1e-06,
    'tessellation_mm': 0.02, 'active_margin_mm': 0.5, 'geometry_device': 'auto', 'geometry_offload': True}
growth: {'enabled': True, 'phase': 0, 'tol': 0.001, 'patience': None, 'quantiles': [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8,
    0.9, 1.0], 'step_tol': 0.0001, 'max_iters_per_step': 25, 'min_iters_per_step': 2, 'samples': 16}
objective: {'type': 'weighted', 'constraints': [], 'constraint_scale': 0.01, 'feasibility_tol': 0.001, 'w_stiffness': 0.5,
    'w_capacity': 0.5, 'min_gain': 0.02, 'floor_stiffness': 1.0, 'floor_capacity': 1.0, 'floor_strength': 1.0,
    'floor_buckling': 1.0, 'floors_as_constraints': True}
export cfg: {'h_mm': 0.2, 'device': 'gpu', 'closed_thr': 0.9, 'escape_holes': True}
```

L-bracket: supports (blue, all DOF fixed; 3 patches) and loads (orange faces; 1 red arrow, one per loaded patch at its area centroid, length and label = its share of the total 200 N). Faces read from bc_faces.npz (as analysed).

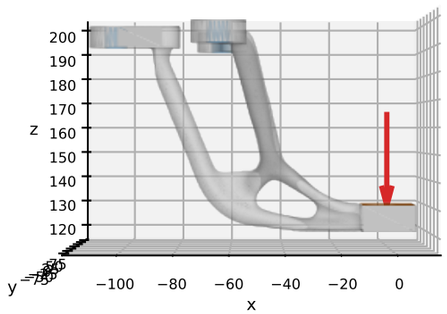
isometric



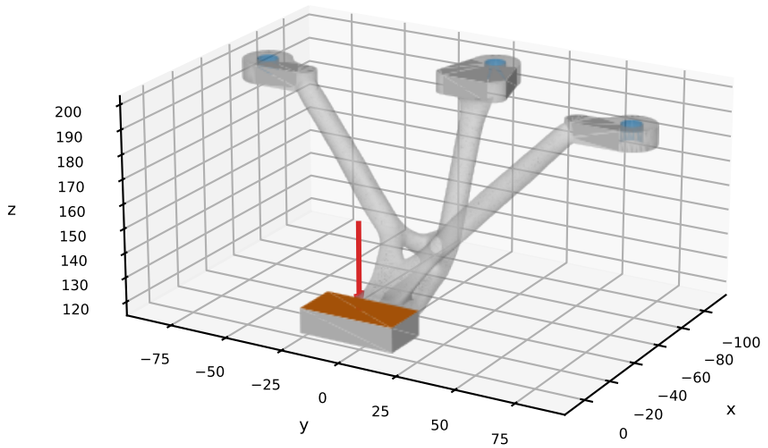
top (+z)



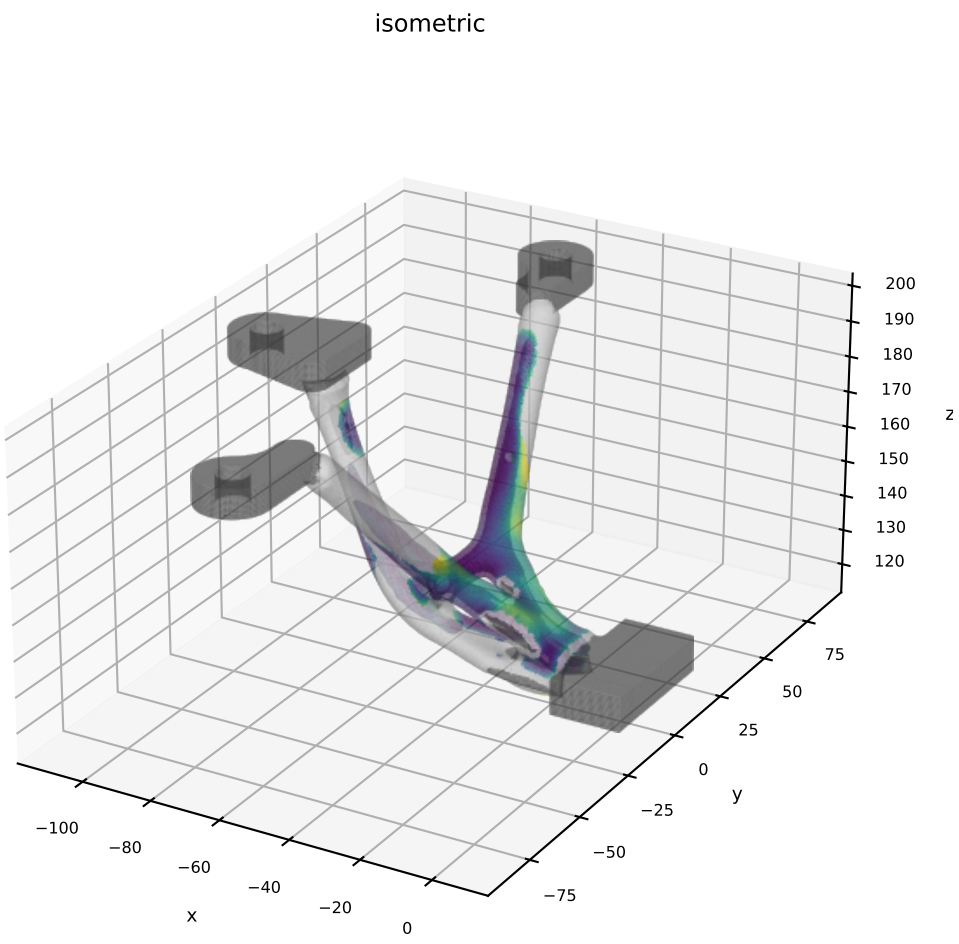
front (-y)

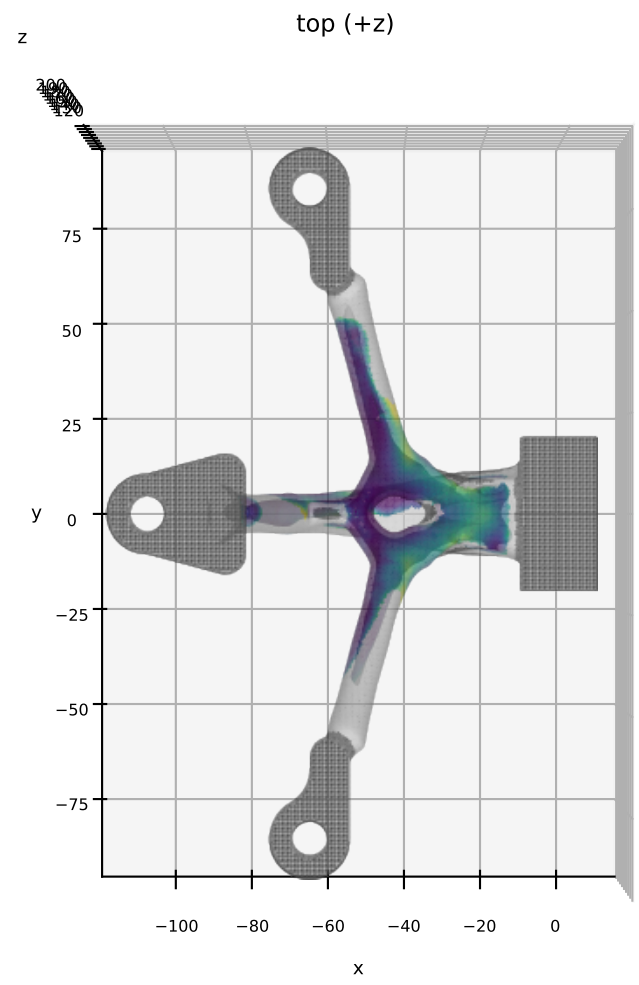


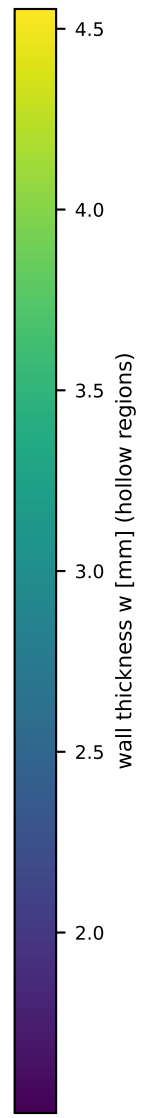
rear isometric

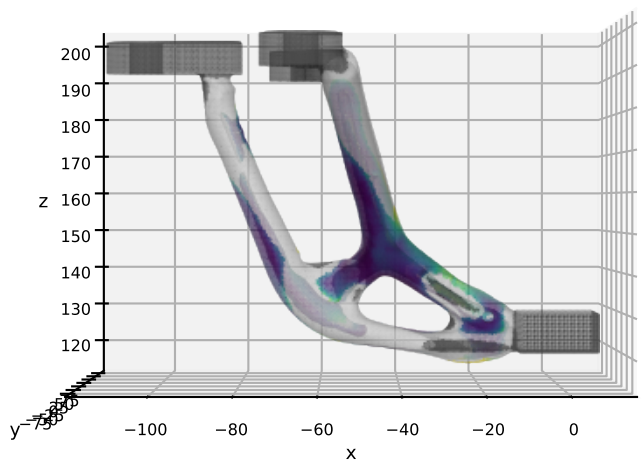


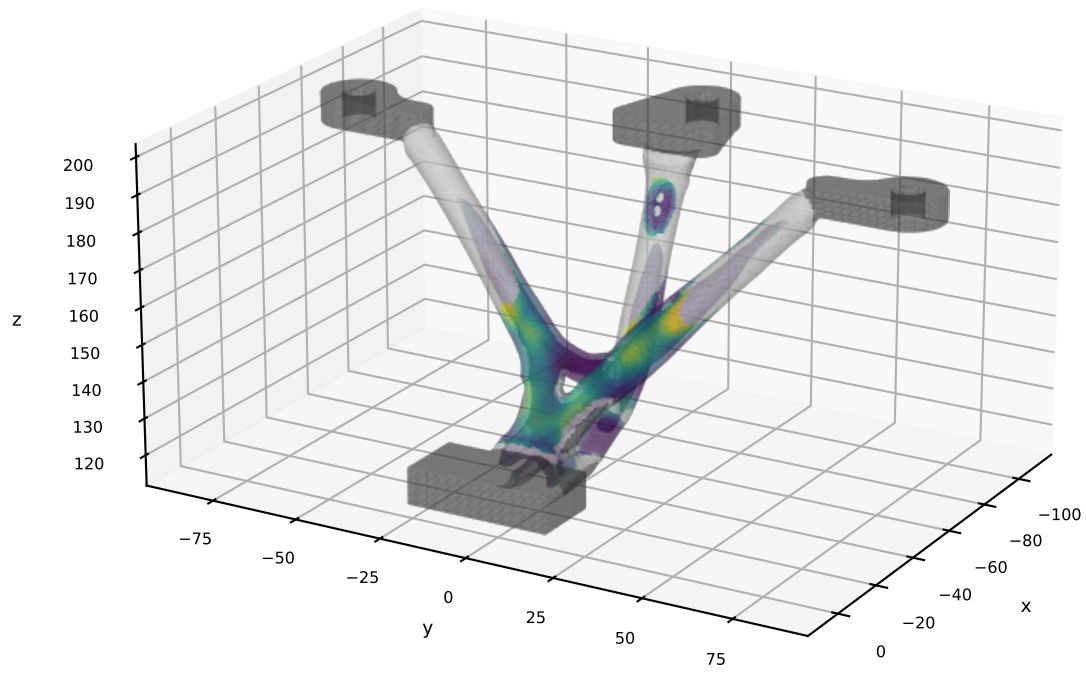
L-bracket: final design (STL 1,599,944 faces, shown decimated to 159,018; wall thickness from fields.vtp)





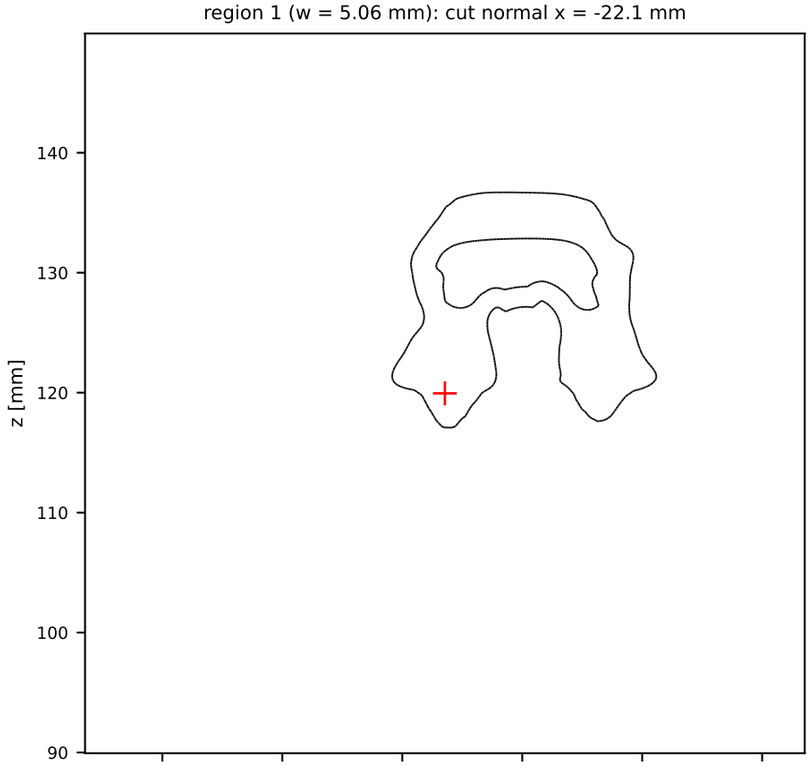


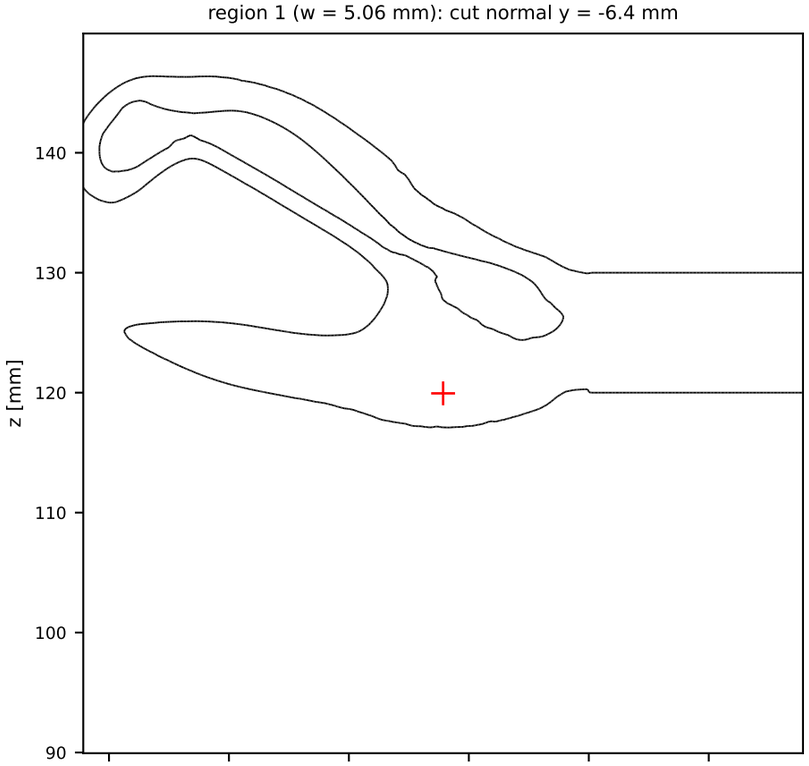


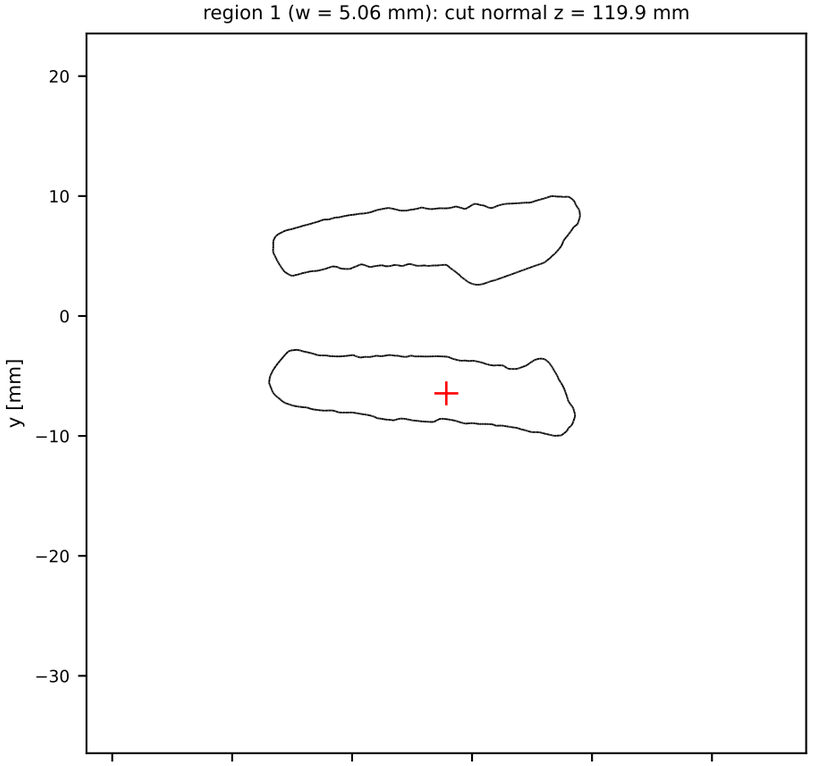


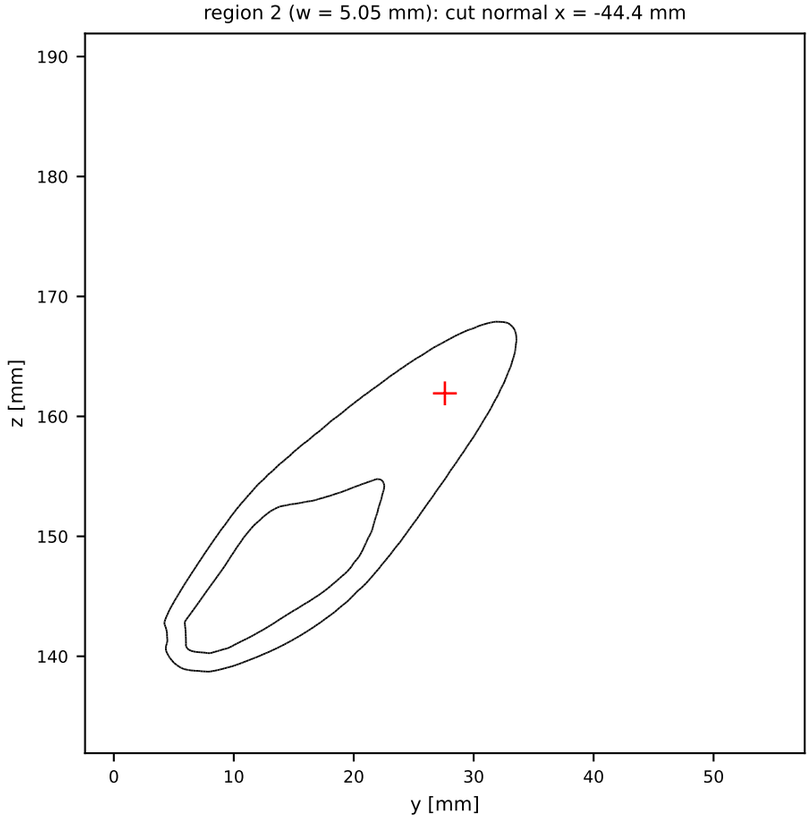
light grey = solid (closed) regions; dark grey = mounts; the colourbar covers the hollow regions only

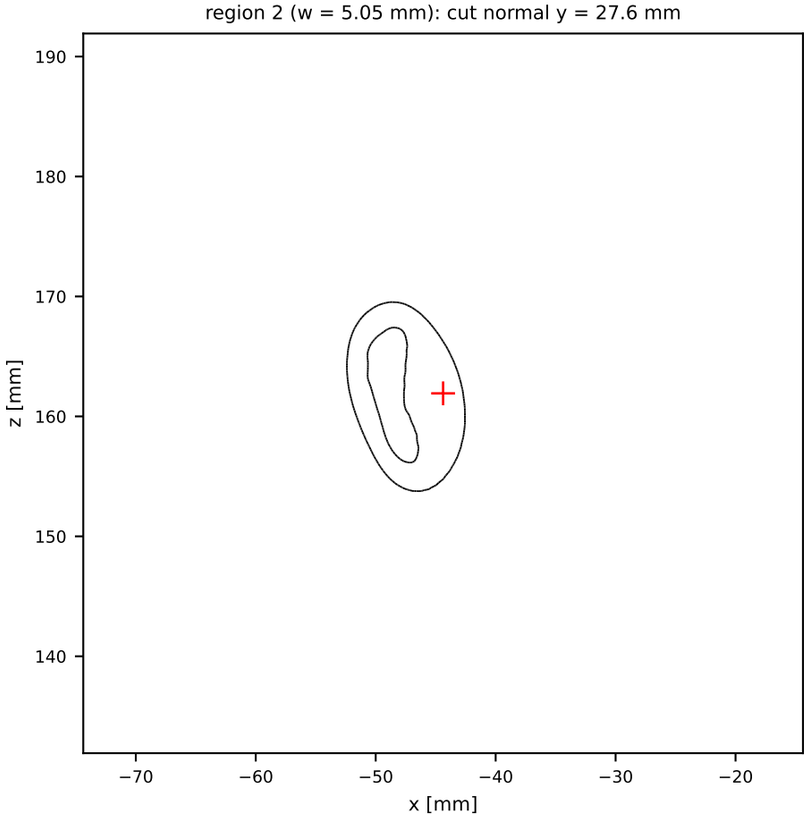
L-bracket: sections of the final design through the two thickest-wall regions (red +); +-30 mm window

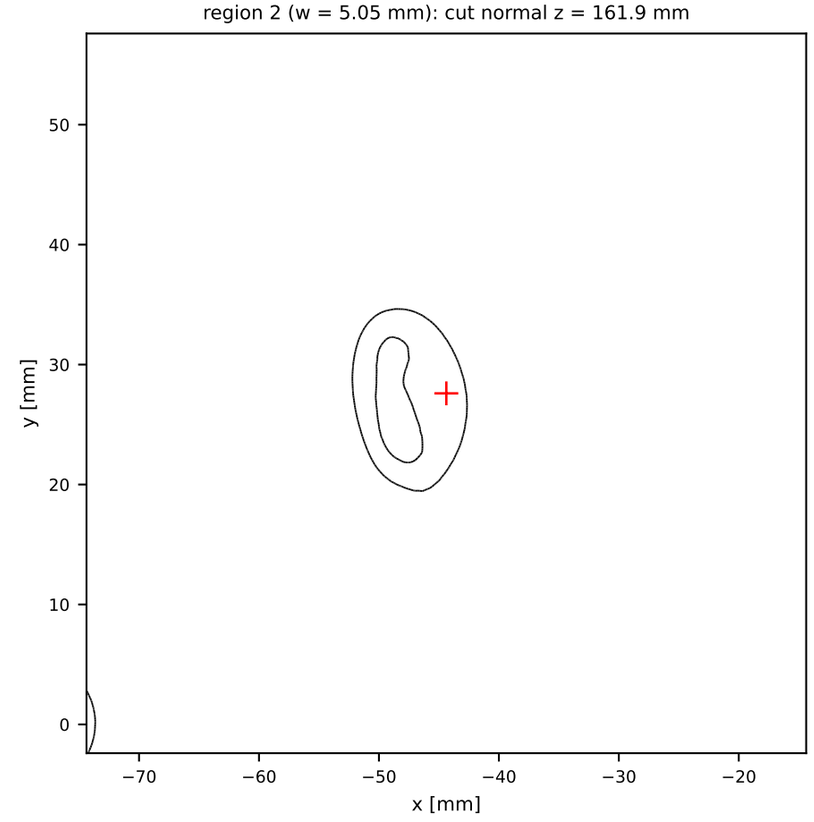






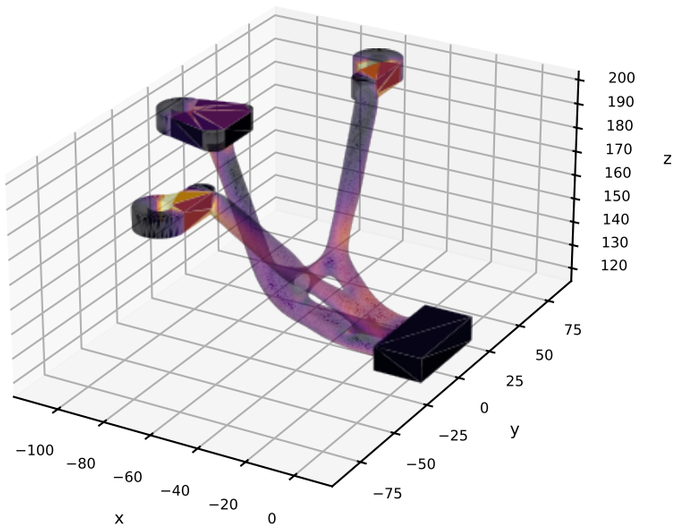




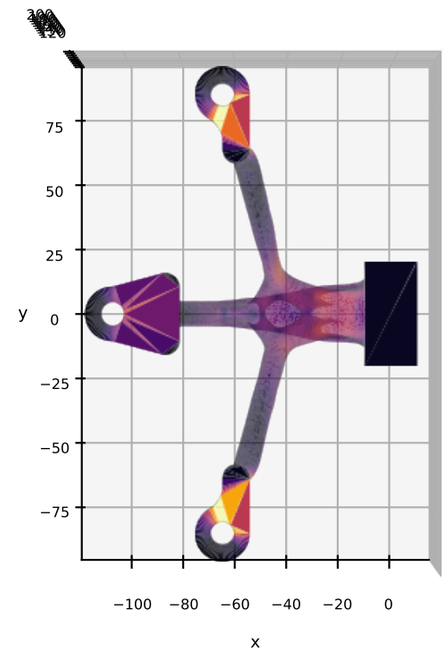


L-bracket: von Mises stress, original solid vs final design (peak solid 42.6, final 41.0 MPa; values from fields.vtp, nearest point)

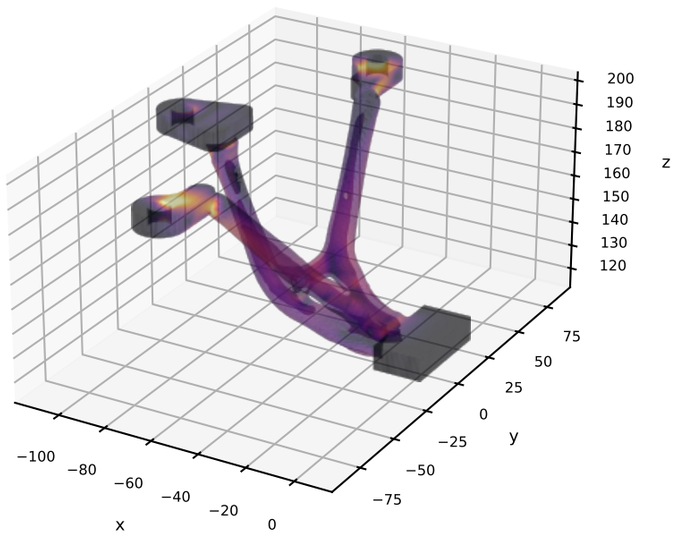
solid: isometric



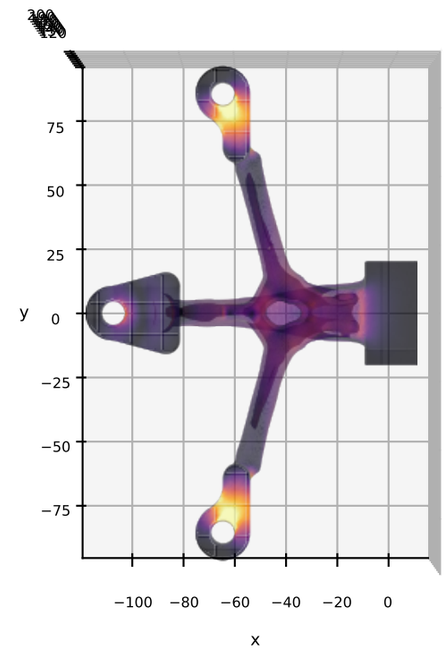
solid: top (+z)

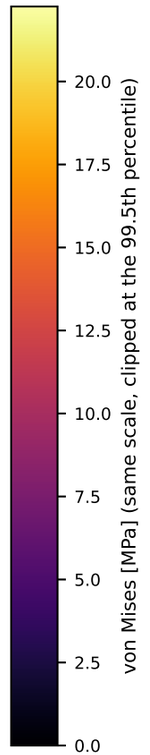


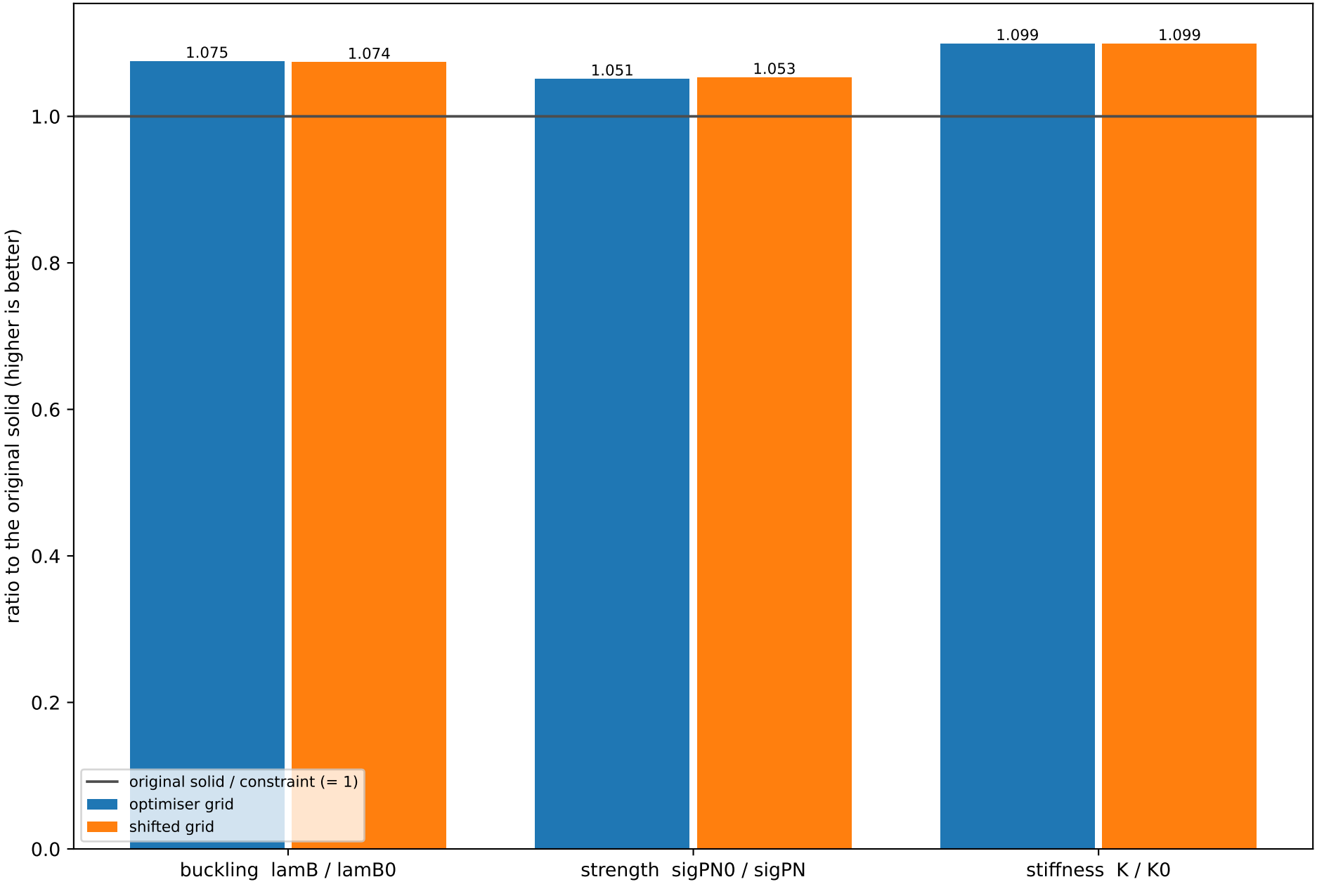
final design: isometric

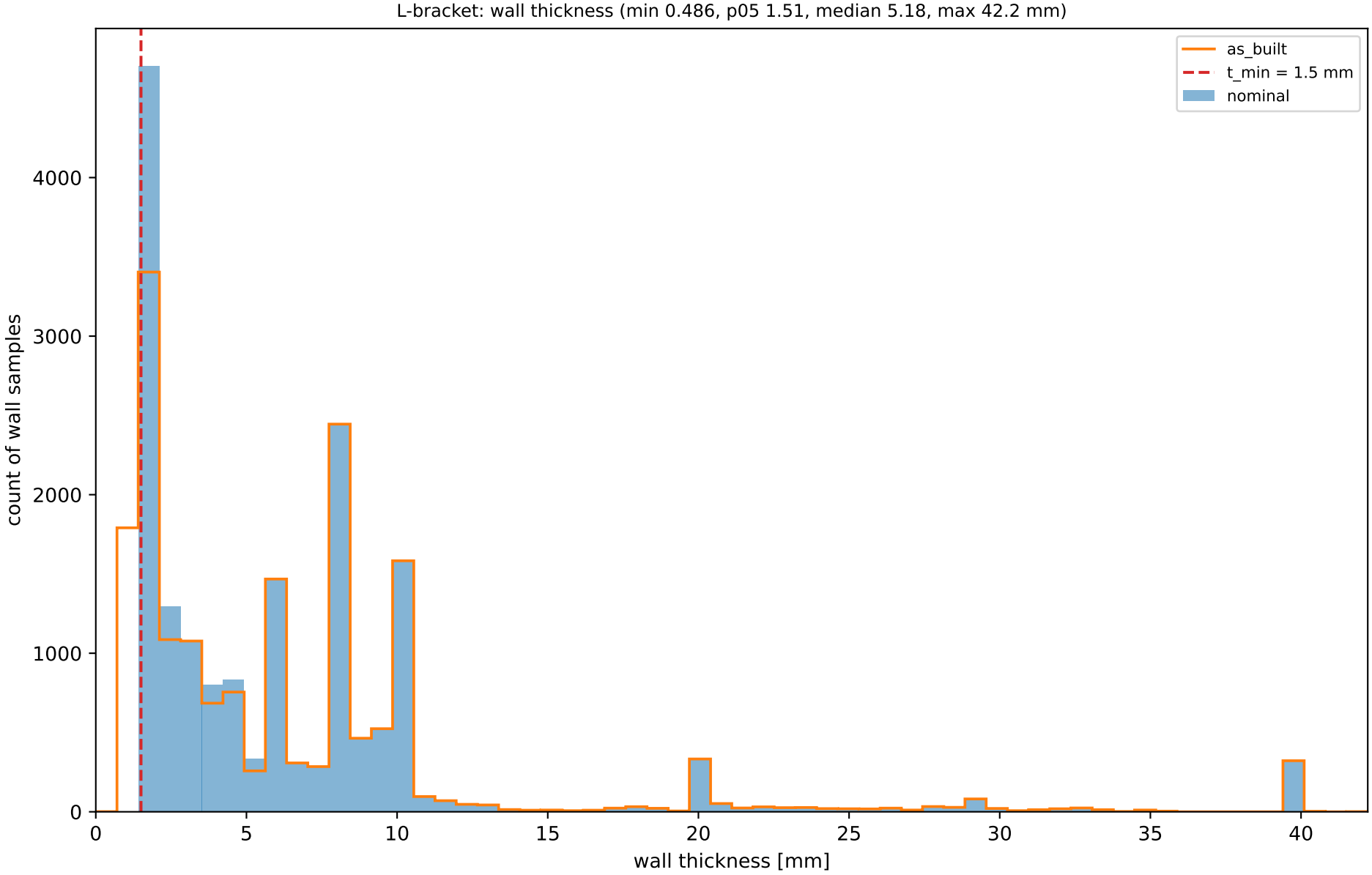


final design: top (+z)



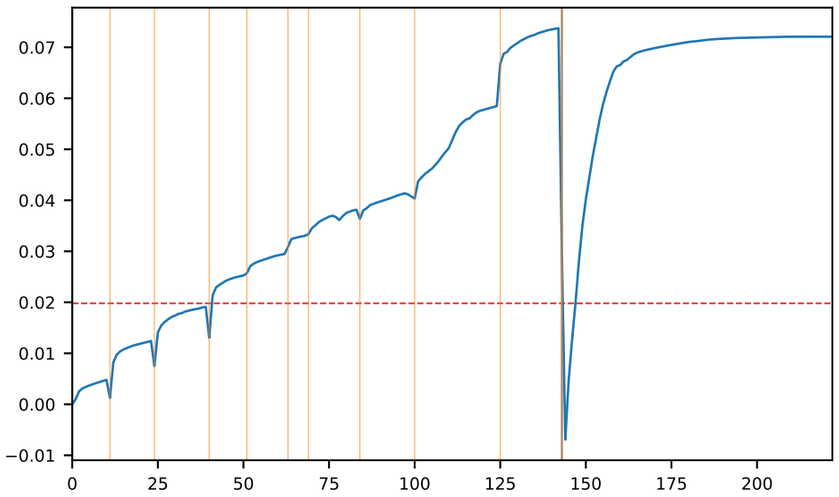



L-bracket: capacity by mode against the solid's constraints. The buckling and strength constraints require both bars >= 1.
ratio to the original solid (higher is better)
0.0
0.2
0.4
0.6
0.8
1.0
1.075
1.074
1.051
1.053
1.099
1.099
original solid / constraint (= 1)
optimiser grid
shifted grid
buckling lamB / lamB0
strength sigPN0 / sigPN
stiffness K / K0

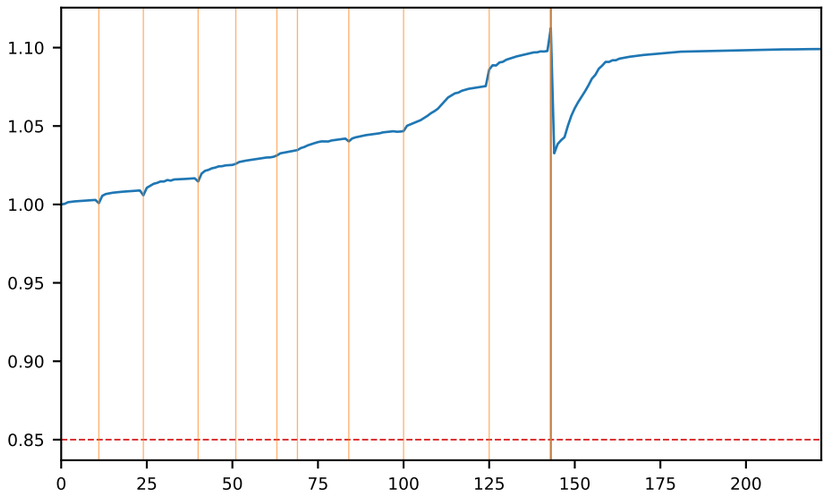

L-bracket: wall thickness (min 0.486, p05 1.51, median 5.18, max 42.2 mm)
as_built
t_min = 1.5 mm
nominal
count of wall samples
wall thickness [mm]
0
1000
2000
3000
4000
0
5
10
15
20
25
30
35
40

L-bracket: convergence history

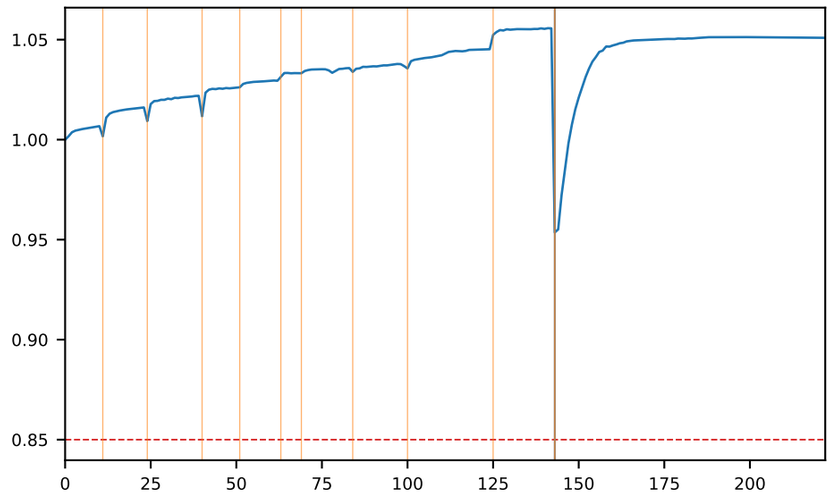
J = weighted objective (accept above ln 1.02)



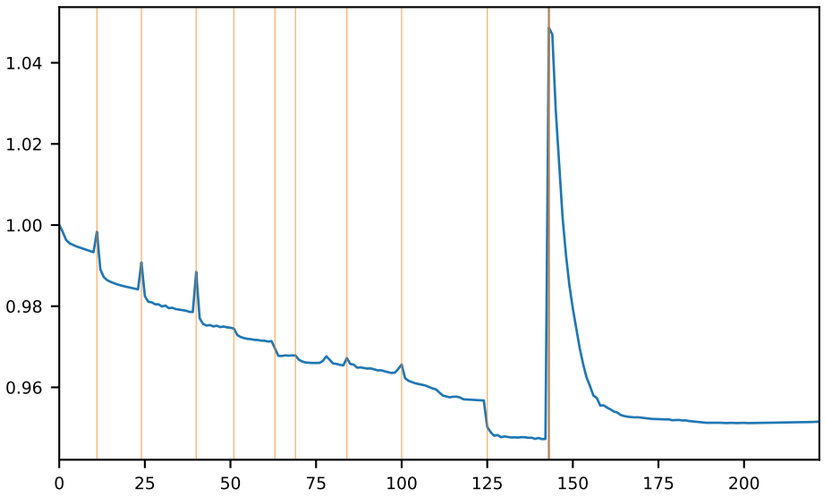
K / K0 (floor 0.85)



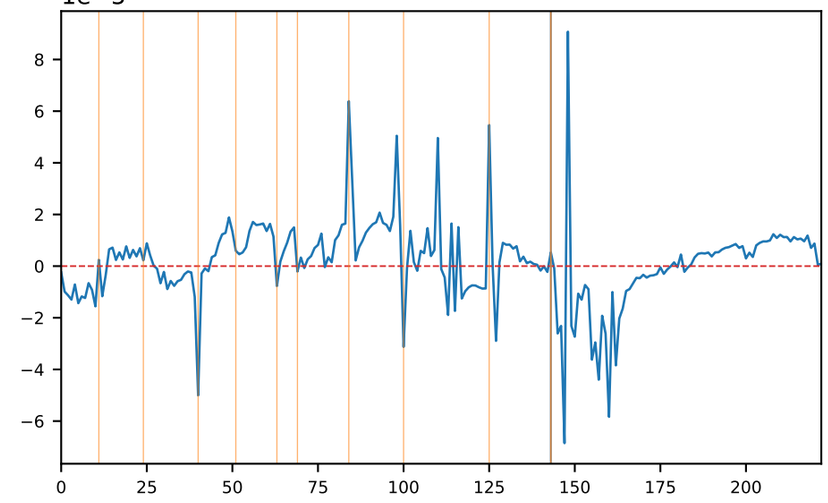
P / P0 (floor 0.85)



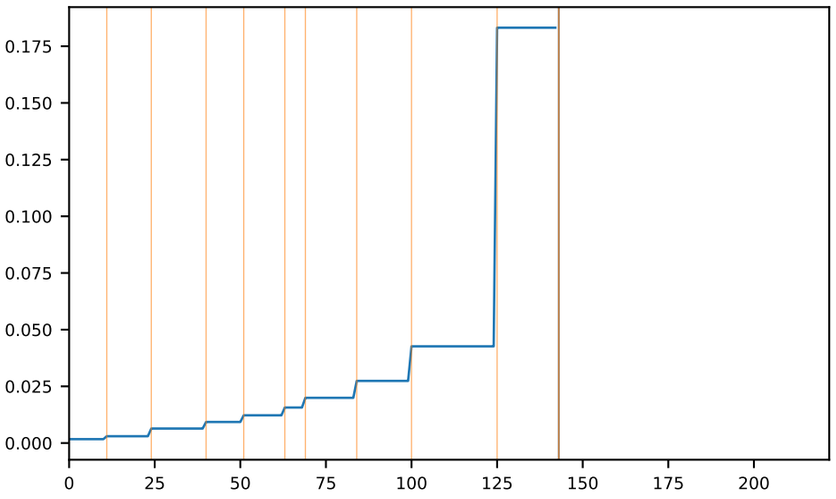
sigPN / sigPN0



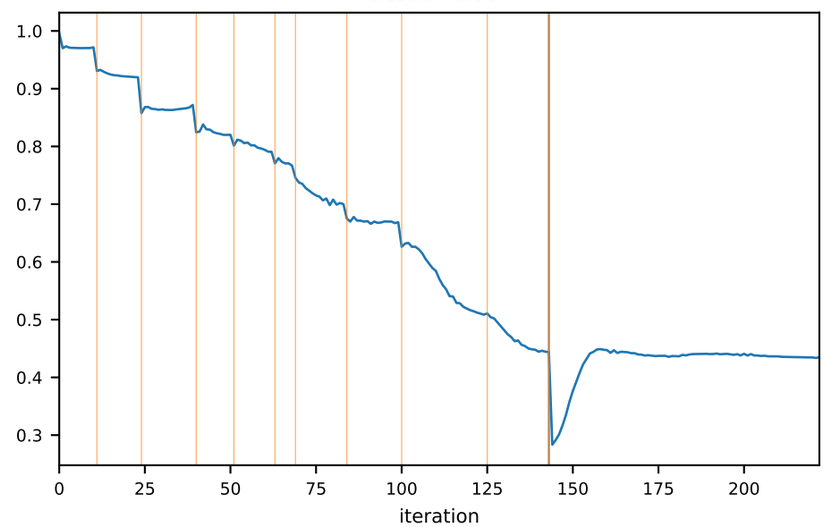
mass error V/V0 - 1



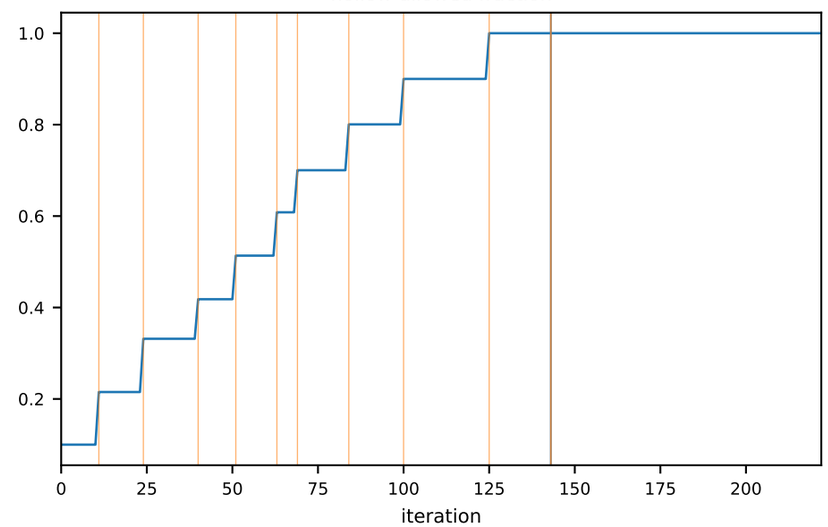
R_k (hollow allowed where R <= R_k)



closed fraction



hollow-allowed fraction

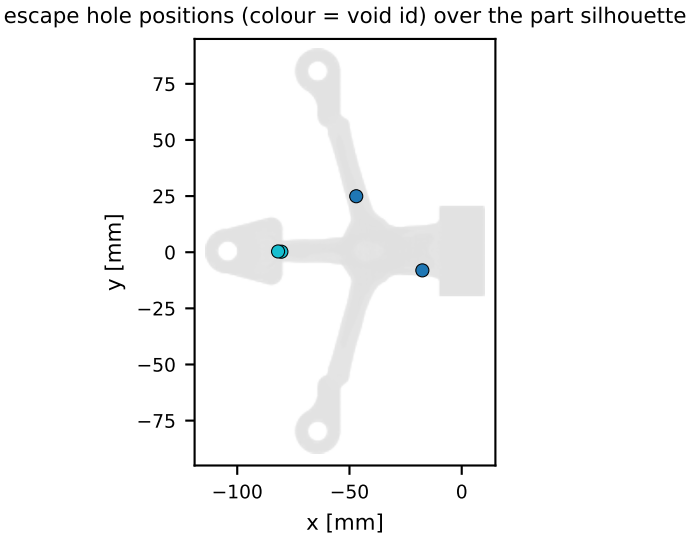


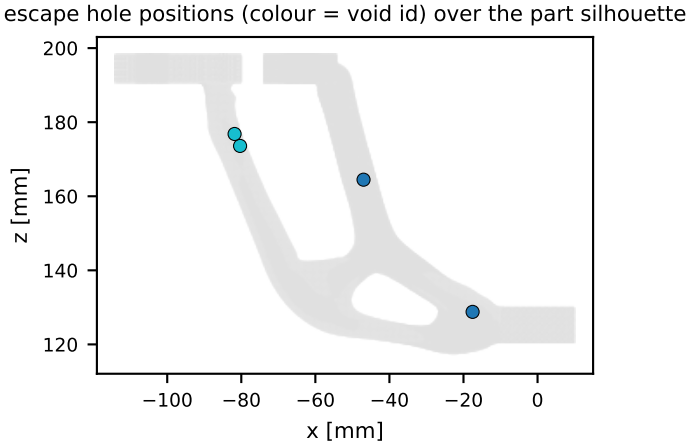
grey vertical: phase boundary
orange vertical: growth step (R_k rise)
iterations 223; growth steps 10
red dashed: constraint / acceptance level

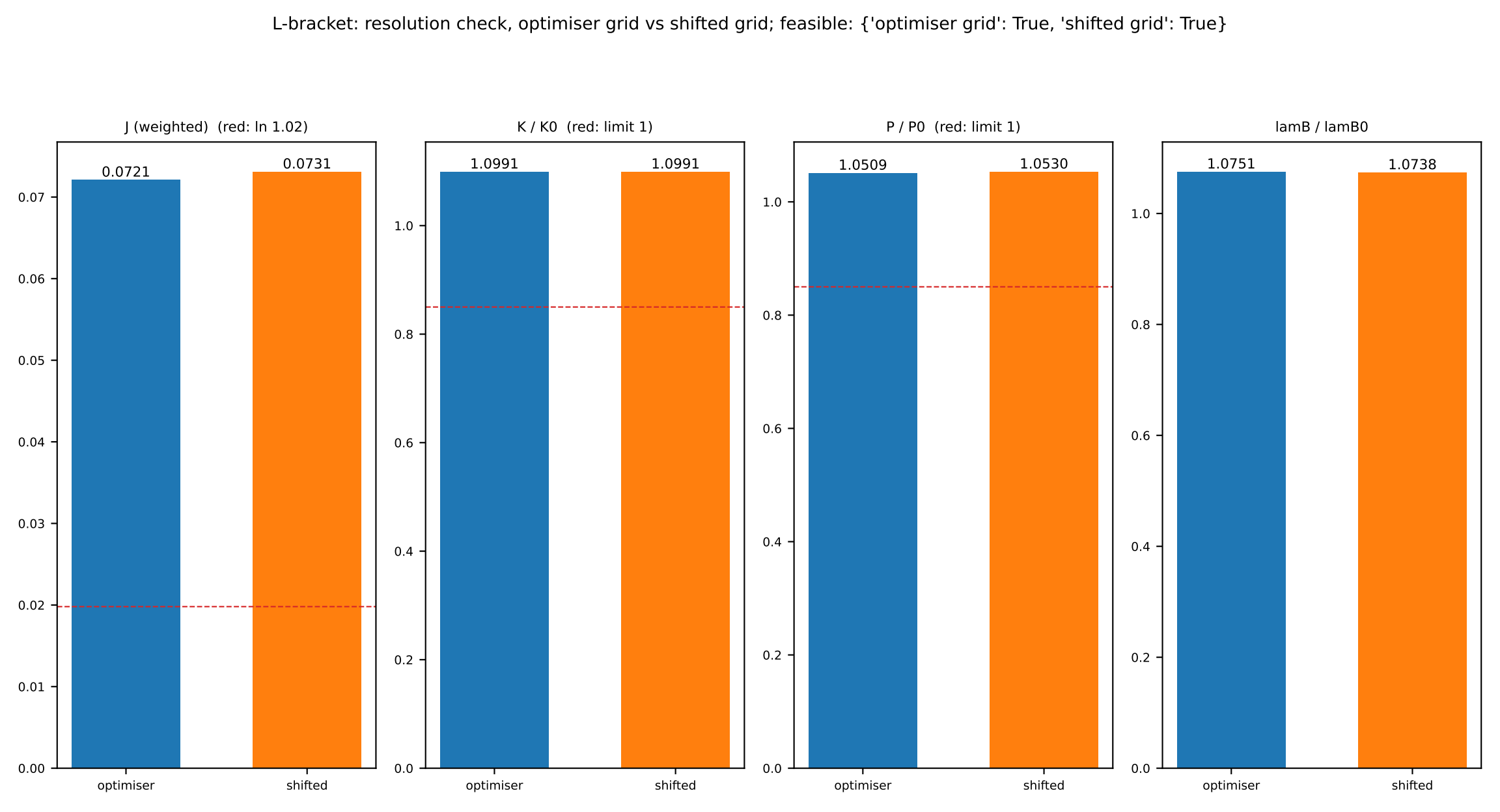
```
enclosed voids: 2;  escape holes: 4;  voids without a hole: 0
  void 0: volume 5024.8 mm3, holes 2
  void 1: volume 87.936 mm3, holes 2
```

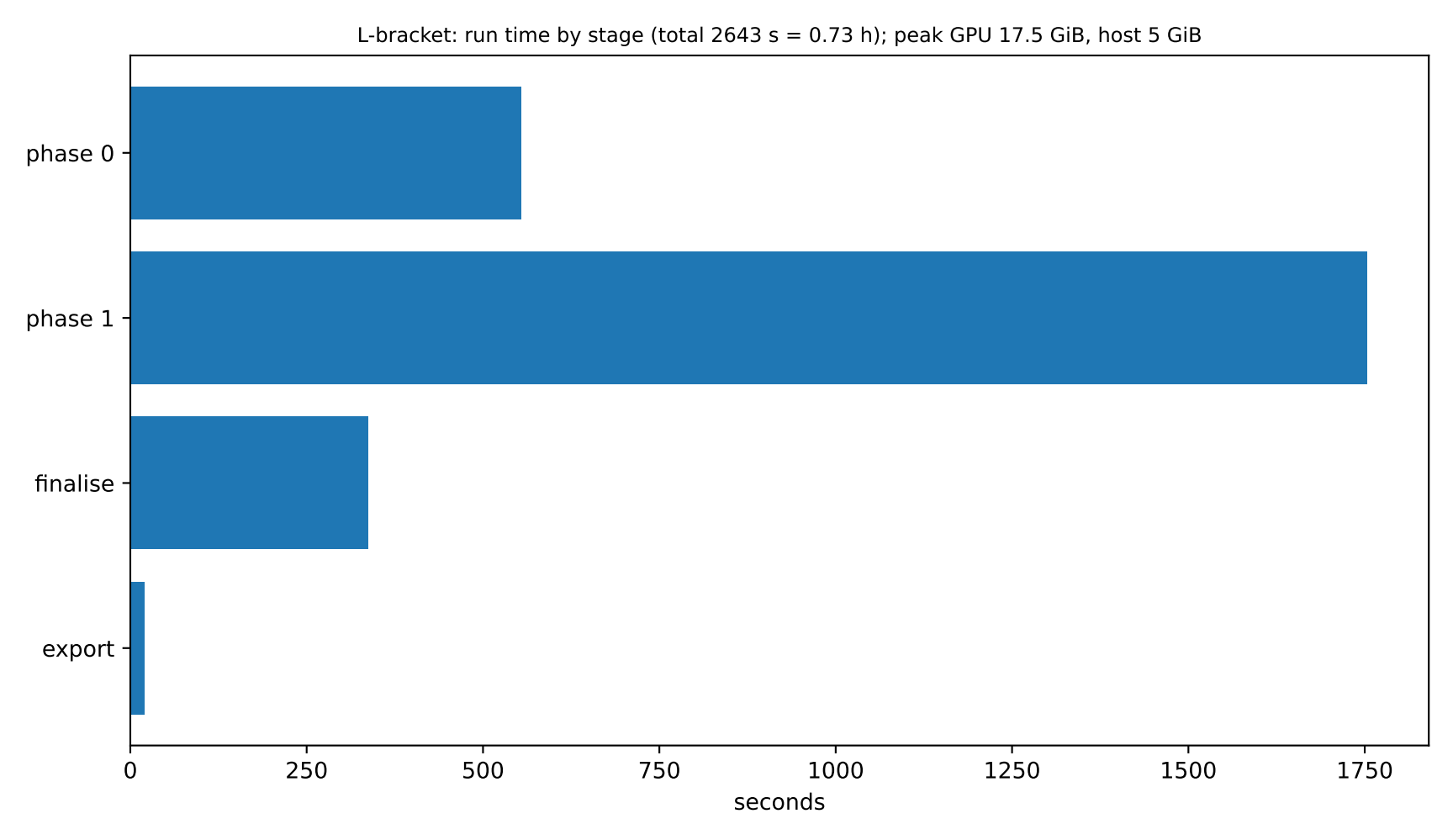
void_id	hole	x	y	z	wall_mm	score_kind
0	0	-46.99	24.93	164.47	1.84	inset_a
0	1	-17.54	-8.08	128.78	1.70	inset_a
1	0	-80.33	0.23	173.59	1.74	inset_a
1	1	-81.81	0.34	176.82	1.98	inset_a

Escape holes sit at the lowest-stress wall points of each void (outward direction in escape_holes.csv).






L-bracket: resolution check, optimiser grid vs shifted grid; feasible: {'optimiser grid': True, 'shifted grid': True}
J (weighted) (red: ln 1.02)
0.0721
0.0731
K / K0 (red: limit 1)
1.0991
1.0991
P / P0 (red: limit 1)
1.0509
1.0530
lamB / lamB0
1.0751
1.0738
optimiser
shifted


L-bracket: run time by stage (total 2643 s = 0.73 h); peak GPU 17.5 GiB, host 5 GiB
phase 0
phase 1
finalise
export
0
250
500
750
1000
1250
1500
1750
seconds